talking about. For the Earth, it's typically about ten million years. So mountains, volcanic and otherwise, must be built on the same timescale; otherwise the Earth would be everywhere smooth as Kansas.*

Volcanic explosions can punch vast quantities of matter—mainly fine droplets of sulfuric acid—into the stratosphere. There, for a year or two, they reflect sunlight back to space and cool the Earth. This happened recently with the Philippine volcano, Mt. Pinatubo, and disastrously in 1815–16 after the eruption of the Indonesian volcano Mt. Tambora, which resulted in the famine-ridden "year without a summer." A volcanic eruption in Taupo, New Zealand, in the year 177 cooled the climate of the Mediterranean, half a world away, and dropped fine particles onto the Greenland ice cap. The explosion of Mt. Mazama in Oregon (which left the caldera now called Crater Lake) in 4803 B.C. had climatic consequences throughout the northern hemisphere. Studies of volcanic effects on the climate were on the investigative path that eventually led to the discovery of nuclear winter. They provide important tests of our use of computer models to predict future climate change. Volcanic particles injected into the upper air are also an additional cause of thinning of the ozone layer.

So a large volcanic explosion in some unfrequented and obscure part of the world can alter the environment on a global scale. Both in their origins and in their effects, volcanos remind us of how vulnerable we are to minor burps and sneezes in the Earth's internal metabolism, and how important it is for us to understand how this subterranean heat engine works.

IN THE FINAL STAGES of formation of the Earth—as well as the Moon, Mars, and Venus—impacts by small worlds are thought to have generated global magma oceans. Molten rock flooded the pre-existing topography. Great floods, tidal waves kilometers high, of flowing, red-hot liquid magma welled up from the interior and poured over the surface of the planet, burying everything in their path: mountains, channels, craters, perhaps even the last evidence of